

# Sebastian "Sebo" Diaz

sebodiaz@csail.mit.edu · sdd@mit.edu · seboddiaz@gmail.com · sebodiaz.github.io

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## OVERVIEW

- HST MIT PhD candidate in RLE & CSAIL.
- Research interests in machine learning, (convex) optimization, representation learning, generalization, data augmentation, and reinforcement learning.

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## EDUCATION

|                                                      |                            |
|------------------------------------------------------|----------------------------|
| Massachusetts Institute of Technology                | 2023 – May 2028 (expected) |
| Ph.D. HST (Computer Science)                         | GPA: 5.00 / 5.00           |
| University of Arizona                                | 2019 – 2023                |
| B.S. in Biomedical Engineering; Minor in Mathematics | GPA: 3.95 / 4.00           |

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## INDUSTRY EXPERIENCE

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|---------------------------------------------------------------|-----------------------|
| Machine Learning Research, Stealth Startup                    | May 2026 – Sept. 2026 |
| Kendall Square, Cambridge, MA                                 |                       |
| • Post-training V/LLM research in non-verifiable RL settings. |                       |

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## FELLOWSHIPS & AWARDS

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|--------------------------------------------------------------------|------|
| Graduate Research Fellowship Program, National Science Foundation  | 2024 |
| MathWorks Graduate Fellow, MathWorks                               | 2024 |
| School of Engineering – Distinguished Graduate Student Fellow, MIT | 2023 |
| Program for Exemplary Mentoring Fellow, MIT                        | 2023 |
| Goldwater STEM Scholarship, Goldwater Foundation                   | 2022 |
| Sophomore of the Year, University of Arizona                       | 2021 |
| Native Health Care Scholar, Udall Foundation                       | 2021 |

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## PUBLICATIONS

### Lead Author

1. **Sebastian Diaz**, Polina Golland, Elfar Adalsteinsson, Neel Dey. “Why Invariance is Not Enough for Biomedical Domain Generalization and How to Fix It”. *Preprint: arXiv:2604.02564*, 2026.
2. **Sebastian Diaz**, Benjamin Billot, Neel Dey, Mingxuan Zhang, Esra Abaci-Turk, Ellen Grant, Polina Golland, Elfar Adalsteinsson. “Robust Fetal Pose Estimation across Gestational Ages via Cross-Population Augmentation”. *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025.
3. **Sebastian Diaz**, Yamin Arefeen, Borjan Gagoski, Ellen Grant, Elfar Adalsteinsson. “Design of Novel RF Pulse for Fetal MRI Refocusing Trains Using Rank Factorization (SLfRank) to Reduce SAR and Improve Image Acquisition Efficiency”. *International Society for Magnetic Resonance in Medicine (ISMRM)*, 2022.

### Contributing Author

1. Yingcheng Liu, Peiqi Wang, **Sebastian Diaz**, Benjamin Billot, Esra Abaci-Turk, Ellen Grant, Polina Golland. "Fetuses Made Simple: Modeling and Tracking of Fetal Shape and Pose". *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025.
2. Molin Zhang, Nicolas Arango, **Sebastian Diaz**, Jacob White, Elfar Adalsteinsson. "Stochastic-Offset-Strategy Enhanced RF Pulse Optimization with Auto-Differentiation". *International Society for Magnetic Resonance in Medicine (ISMRM)*, 2024.
3. Ricky Cordova, Kelli Kiekens, Susan Burrell, William Drake, Zaynah Kmeid, Photini Rice, Andrew Rocha, **Sebastian Diaz**, Shigehiro Yamada, Michael Yozwiak, Omar L. Nelson, Gustavo C. Rodriguez, John Heusinkveld, Ie-Ming Shih, David S. Alberts, Jennifer K. Barton. "Sub-Millimeter Endoscope Demonstrates Feasibility of In Vivo Reflectance Imaging, Fluorescence Imaging, and Cell Collection in the Fallopian Tubes". *Journal of Biomedical Optics*, 2021.

INVITED TALKS

|                                                                                               |                |
|-----------------------------------------------------------------------------------------------|----------------|
| <b>RepreSeg: Stable Representations for Domain Generalized Segmentation</b>                   |                |
| MIT IMES Seminar, Cambridge, MA                                                               | Dec. 12, 2025  |
| <b>Learning Alongside Invariant Representations: Robust Generalization in Medical Imaging</b> |                |
| MIT IMES Department Retreat, Boston, MA                                                       | Sept. 29, 2025 |

RELEVANT COURSEWORK

|                        |                                                                             |
|------------------------|-----------------------------------------------------------------------------|
| MIT                    | Deep Learning, Numerical Methods, Computer Vision, fMRI Neuroscience        |
| Harvard Medical School | Pathology, Immunology, Genetics, Cardiovascular & Pulmonary Pathophysiology |

SKILLS

|             |                                     |
|-------------|-------------------------------------|
| Programming | Python, PyTorch, C++                |
| Technical   | Numerical optimization, probability |

SERVICE

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|-------------------------|---------|
| Reviewer, MICCAI: PIPPI | 2025–26 |
| Reviewer, CVPR          | 2026    |
| Reviewer, NeurIPS       | 2026    |